

$$1b) \sum_{n=0}^{\infty} \cos(n\theta) x^{4n} \quad \text{ou } x \in \mathbb{R} \text{ et } \theta \in \mathbb{R}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (x^{4n} - x^{4n} \cos 2n\theta)$$

Convergenes: $S_1 = \sum_{n=0}^{\infty} x^{4n} = \begin{cases} \frac{1}{1-x^4} & \text{si } |x| < 1 \\ \text{div} & \text{si } |x| \geq 1 \end{cases}$

$$S_2 = \sum_{n=0}^{\infty} x^{4n} \cos 2n\theta = \sum_{n=0}^{\infty} \operatorname{Re}(x^4 e^{i2n\theta})$$

Si $|x| < 1$, $\sum_{n=0}^{\infty} (x^4 e^{i2n\theta})^n$ converge donc $S_2 \in \mathbb{R}$ et on a

$$S_2 = \operatorname{Re} \left(\sum_{n=0}^{\infty} (x^4 e^{i2n\theta}) \right) = \operatorname{Re} \left(\frac{1}{1 - x^4 e^{i2\theta}} \right) = \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2}$$

Si $|x| \geq 1$, $\frac{1}{x^4} \cos 2n\theta$ diverge donc $S_2 = \frac{1}{1-x^4} + \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2}$

Si $|x| \geq 1$, $(\cos(n\theta) x^{4n}) \not\rightarrow 0$ et car $\cos(n\theta) x^{4n} \geq 0$, $S_2 \rightarrow \infty$

Si $\theta \in \mathbb{R}$

$$\frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2} + \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2}$$

$$= \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2} + \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2}$$

$$= \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2} + \frac{1 - x^4 \cos 2\theta}{(1 - x^4 \cos 2\theta)^2 + (x^4 \sin 2\theta)^2}$$

Soit $a \in]0, \frac{\pi}{2}[$

Posons $g = \sum_{n=0}^{\infty} \ln \cos\left(\frac{a}{2^n}\right)$

$$= \lim_{N \rightarrow +\infty} \ln \left(\prod_{n=0}^N \cos\left(\frac{a}{2^n}\right) \right)$$

Soit $N \in \mathbb{N}^*$. Posons $C_N = \prod_{n=0}^N \cos\left(\frac{a}{2^n}\right)$ et $S_N = \prod_{n=0}^N \sin\left(\frac{a}{2^n}\right)$

On a: $C_N \times S_N = \prod_{n=0}^N \frac{1}{2} \sin\left(\frac{a}{2^{n+1}}\right)$ car $\forall x \in \mathbb{R}, \sin x = 2 \cos x \sin \frac{x}{2}$

$$= \frac{1}{2} \left(\prod_{n=0}^{N-1} \sin\left(\frac{a}{2^{n+1}}\right) \right) \times \sin(2a)$$

d'où $C_N \times S_N = \frac{1}{2} \times \frac{S_N}{\sin\left(\frac{a}{2^N}\right)} \Rightarrow C_N = \frac{1}{2 \sin\left(\frac{a}{2^N}\right)}$ car que $S_N \neq 0$

d'où $g = \lim_{N \rightarrow +\infty} \ln \left(\frac{1}{2 \sin\left(\frac{a}{2^N}\right)} \right)$

Soit $a \in]0, \frac{\pi}{2}[$

Ponons $S = \sum_{n=0}^{+\infty} \ln\left(\cos\left(\frac{a}{2^n}\right)\right)$ qui est bien définie car que

$$0 < \frac{a}{2^n} < \frac{\pi}{2} \quad (\forall n \in \mathbb{N}) \Rightarrow \cos\left(\frac{a}{2^n}\right) > 0$$

$$\text{On a: } S = \lim_{N \rightarrow +\infty} \ln\left(\prod_{n=0}^N \cos\left(\frac{a}{2^n}\right)\right)$$

$$\text{Soit } N \in \mathbb{N}^*. \text{ Ponons } C_N = \prod_{n=0}^N \cos \frac{a}{2^n} \text{ et } S_N = \sum_{n=0}^N \sin \frac{a}{2^n}$$

$$\text{On a: } C_N \times S_N = \prod_{n=0}^N \frac{1}{2} \sin\left(\frac{a}{2^{n+1}}\right)$$

$$= \frac{1}{2^{N+1}} \times \sin(2a) \times \prod_{n=1}^{N+1} \sin\left(\frac{a}{2^{n-1}}\right)$$

$$= \frac{\sin(2a)}{2^{N+1}} \times \prod_{n=0}^{N-1} \sin \frac{a}{2^n}$$

$$\text{d'où } C_N \times S_N = \frac{\sin(2a)}{2^{N+1}} \times \frac{S_N}{\sin\left(\frac{a}{2^N}\right)} \Rightarrow C_N = \frac{\sin(2a)}{2^{N+1} \sin\left(\frac{a}{2^N}\right)}$$

car que $S_N \neq 0$

$$\text{aussi } S = \lim_{N \rightarrow +\infty} \ln\left(\frac{\sin(2a)}{2^{N+1} \sin\left(\frac{a}{2^N}\right)}\right)$$

$$= \lim_{N \rightarrow +\infty} \ln\left(\frac{\sin(2a)}{2a} \times \frac{a/2^N}{\sin(a/2^N)}\right)$$

$$S = \ln\left(\frac{\sin(2a)}{2a}\right) \text{ car que } \ln \text{ est continue en } \frac{\sin(2a)}{2a} \in \mathbb{R}_+^*$$